

Reverse-check review package — SMWeinberg_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	SMWeinberg/model/SMWeinberg_gen.fr
original model name	SMWeinberg_gen (hidden from the agent)
paper	SMWeinberg/text/2012.09882.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LNKin (:=)

$$I/2 \, N1bar[s1] . Ga[v, s1, s2] . del[N1[s2], v] - 1/2 \, mN1 \, N1bar[s1] . N1[s1]$$

LNCCbare (:=)

$$gw/Sqrt[2] * N1bar.W[m] . ProjM[m] . e + gw/Sqrt[2] * N1bar.W[m] . ProjM[m] . mu + gw/Sqrt[2] * N1bar.W[m] . ProjM[m] . ta$$

LNCC (:=)

$$LNCCbare + HC[LNCCbare]$$

LNNCbare (:=)

$$1/2 * gw/cw * N1bar.Z[m] . ProjM[m] . ve + 1/2 * gw/cw * N1bar.Z[m] . ProjM[m] . vm + 1/2 * gw/cw * N1bar.Z[m] . ProjM[m] . vt$$

LNNC (:=)

$$LNNCbare + HC[LNNCbare]$$

LNHbare (:=)

$$- gw*mN1/(2*MW) * (1 + H*gw/(4*MW)) * N1bar.ProjM.ve \, H - gw*mN1/(2*MW) * (1 + H*gw/(4*MW)) * N1bar.ProjM.vt \, H - gw*mN1/(2*MW) * (1 + H*gw/(4*MW)) * N1bar.ProjM.vt \, H$$

LNHX (:=)

$$LNHbare + HC[LNHbare]$$

LNGbare (:=)

```
I *gw*mN1/(2*Sqrt[2]*MW) * (1 + H*gw/(2*MW)) * ebar.ProjP.N1 GPbar + I *gw*mN1/(2*Sqrt[2]*MW)
↪ * (1 + H*gw/(2*MW)) * mubar.ProjP.N1 GPbar + I *gw*mN1/(2*Sqrt[2]*MW) * (1 + H*gw/(2*MW))
↪ * tabar.ProjP.N1 GPbar + I *gw*mN1/(2*Sqrt[2]*MW) * (1 + H*gw/(2*MW)) *
↪ N1bar.ProjP.CC[e] GPbar + I *gw*mN1/(2*Sqrt[2]*MW) * (1 + H*gw/(2*MW)) *
↪ N1bar.ProjP.CC[mu] GPbar + I *gw*mN1/(2*Sqrt[2]*MW) * (1 + H*gw/(2*MW)) *
↪ N1bar.ProjP.CC[ta] GPbar + I *gw*mN1/(2*MW) * (1 + H*gw/(2*MW)) * vbar.ProjP.N1 GO + I
↪ *gw*mN1/(2*MW) * (1 + H*gw/(2*MW)) * vmbar.ProjP.N1 GO + I *gw*mN1/(2*MW) * (1 +
↪ H*gw/(2*MW)) * vtbar.ProjP.N1 GO
```

LNGX (:=)

LNGbare + HC[LNGbare]

LNGGbare (:=)

```
gw*gw*mN1/(4*MW*MW) * ebar.ProjP.CC[e] GPbar GPbar + gw*gw*mN1/(4*MW*MW) * mubar.ProjP.CC[e]
↪ GPbar GPbar + gw*gw*mN1/(4*MW*MW) * tabar.ProjP.CC[e] GPbar GPbar + gw*gw*mN1/(4*MW*MW)
↪ * ebar.ProjP.CC[mu] GPbar GPbar + gw*gw*mN1/(4*MW*MW) * mubar.ProjP.CC[mu] GPbar GPbar +
↪ gw*gw*mN1/(4*MW*MW) * tabar.ProjP.CC[mu] GPbar GPbar + gw*gw*mN1/(4*MW*MW) *
↪ ebar.ProjP.CC[ta] GPbar GPbar + gw*gw*mN1/(4*MW*MW) * mubar.ProjP.CC[ta] GPbar GPbar +
↪ gw*gw*mN1/(4*MW*MW) * tabar.ProjP.CC[ta] GPbar GPbar + gw*gw*mN1/(8*MW*MW) *
↪ vbar.ProjP.N1 GO GO + gw*gw*mN1/(8*MW*MW) * vmbar.ProjP.N1 GO GO + gw*gw*mN1/(8*MW*MW)
↪ * vtbar.ProjP.N1 GO GO + gw*gw*mN1/(4*Sqrt[2]*MW*MW) * ebar.ProjP.N1 GPbar GO +
↪ gw*gw*mN1/(4*Sqrt[2]*MW*MW) * mubar.ProjP.N1 GPbar GO + gw*gw*mN1/(4*Sqrt[2]*MW*MW) *
↪ tabar.ProjP.N1 GPbar GO + gw*gw*mN1/(4*Sqrt[2]*MW*MW) * N1bar.ProjP.CC[e] GPbar GO +
↪ gw*gw*mN1/(4*Sqrt[2]*MW*MW) * N1bar.ProjP.CC[mu] GPbar GO + gw*gw*mN1/(4*Sqrt[2]*MW*MW)
↪ * N1bar.ProjP.CC[ta] GPbar GO
```

LNGGX (:=)

LNGGbare + HC[LNGGbare]

LD5 (:=)

LNKin + LNCC + LNNC + LNHX + LNGX + LNGGX

LFull (:=)

LSM + LD5

Blank-slate reconstruction

Reconstructed Physics From sanitized.fr

Conventions

Let

$$\ell_i = (e, \mu, \tau), \quad \nu_i = (\nu_e, \nu_\mu, \nu_\tau), \quad i, j \in \{e, \mu, \tau\}.$$

The new fermion is denoted $N \equiv N1$. It is self-conjugate:

$$N^c = N.$$

FeynRules projectors are translated as

$$\text{ProjM} = P_L = \frac{1 - \gamma^5}{2}, \quad \text{ProjP} = P_R = \frac{1 + \gamma^5}{2}.$$

The charged fields are interpreted by charge conservation as

$$\mathbf{W} = W^+, \quad \mathbf{GPbar} = G^-,$$

with the hermitian conjugate terms containing W^- and G^+ . The neutral Goldstone is

$$\mathbf{G0} = G^0.$$

No DC covariant derivative appears in the new-sector kinetic term. The file uses `del`, so the kinetic derivative is the ordinary derivative:

$$D_\mu N = \partial_\mu N.$$

This is consistent with the declared new field carrying no color, weak-isospin, or hypercharge indices and electric charge $Q = 0$. Thus the new kinetic term contains no gluon, W , B , photon, or Z gauge connection.

The internal mass parameter is

$$m_N \equiv \mathbf{mN1} = \frac{v^2}{\Lambda} |C_{ee} + C_{e\mu} + C_{e\tau} + C_{\mu\mu} + C_{\mu\tau} + C_{\tau\tau}|.$$

Lagrangian

LNKin

$$\mathcal{L}_{\mathbf{LNKin}} = \frac{i}{2} \bar{N} \gamma^\mu \partial_\mu N - \frac{1}{2} m_N \bar{N} N$$

LNCCbare

$$\mathcal{L}_{\mathbf{LNCCbare}} = \frac{g_w}{\sqrt{2}} \sum_i \bar{N} \gamma^\mu P_L \ell_i W_\mu^+$$

LNCC

$$\mathcal{L}_{\mathbf{LNCC}} = \frac{g_w}{\sqrt{2}} \sum_i (\bar{N} \gamma^\mu P_L \ell_i W_\mu^+ + \bar{\ell}_i \gamma^\mu P_L N W_\mu^-)$$

LNNCbare

$$\mathcal{L}_{\mathbf{LNNCbare}} = \frac{g_w}{2c_w} \sum_i \bar{N} \gamma^\mu P_L \nu_i Z_\mu$$

LNNC

$$\mathcal{L}_{\mathbf{LNNC}} = \frac{g_w}{2c_w} \sum_i (\bar{N} \gamma^\mu P_L \nu_i + \bar{\nu}_i \gamma^\mu P_L N) Z_\mu$$

LNHbare

$$\mathcal{L}_{\mathbf{LNHbare}} = -\frac{g_w m_N}{2M_W} \left(1 + \frac{g_w H}{4M_W} \right) H \sum_i \bar{N} P_L \nu_i$$

Equivalently, expanded in powers of H ,

$$\mathcal{L}_{\mathbf{LNHbare}} = -\frac{g_w m_N}{2M_W} H \sum_i \bar{N} P_L \nu_i - \frac{g_w^2 m_N}{8M_W^2} H^2 \sum_i \bar{N} P_L \nu_i.$$

LNHX

$$\mathcal{L}_{\text{LNHX}} = -\frac{g_w m_N}{2M_W} \left(1 + \frac{g_w H}{4M_W}\right) H \sum_i (\bar{N} P_L \nu_i + \bar{\nu}_i P_R N)$$

LNGbare

$$\begin{aligned} \mathcal{L}_{\text{LNGbare}} = & \frac{ig_w m_N}{2\sqrt{2}M_W} \left(1 + \frac{g_w H}{2M_W}\right) G^- \sum_i (\bar{\ell}_i P_R N + \bar{N} P_R \ell_i^c) \\ & + \frac{ig_w m_N}{2M_W} \left(1 + \frac{g_w H}{2M_W}\right) G^0 \sum_i \bar{\nu}_i P_R N. \end{aligned}$$

LNGX

$$\mathcal{L}_{\text{LNGX}} = \mathcal{L}_{\text{LNGbare}} + \mathcal{L}_{\text{LNGbare}}^\dagger$$

Explicitly, this adds the G^+ conjugates and the conjugate neutral-Goldstone bilinears.

LNGGbare

$$\begin{aligned} \mathcal{L}_{\text{LNGGbare}} = & \frac{g_w^2 m_N}{4M_W^2} (G^-)^2 \sum_{i,j} \bar{\ell}_i P_R \ell_j^c \\ & + \frac{g_w^2 m_N}{8M_W^2} (G^0)^2 \sum_i \bar{\nu}_i P_R N \\ & + \frac{g_w^2 m_N}{4\sqrt{2}M_W^2} G^- G^0 \sum_i (\bar{\ell}_i P_R N + \bar{N} P_R \ell_i^c). \end{aligned}$$

The first sum is over all nine ordered charged-lepton flavor combinations:

$$(i, j) = (e, e), (\mu, e), (\tau, e), (e, \mu), (\mu, \mu), (\tau, \mu), (e, \tau), (\mu, \tau), (\tau, \tau).$$

LNGGX

$$\mathcal{L}_{\text{LNGGX}} = \mathcal{L}_{\text{LNGGbare}} + \mathcal{L}_{\text{LNGGbare}}^\dagger$$

LD5

$$\mathcal{L}_{\text{LD5}} = \mathcal{L}_{\text{LNKin}} + \mathcal{L}_{\text{LNCC}} + \mathcal{L}_{\text{LNNC}} + \mathcal{L}_{\text{LNHX}} + \mathcal{L}_{\text{LNGX}} + \mathcal{L}_{\text{LNGGX}}$$

LFull

$$\mathcal{L}_{\text{LFull}} = \mathcal{L}_{\text{SM}} + \mathcal{L}_{\text{LD5}}$$

The file does not define **LSM**; it only adds the new-sector terms above to the Standard Model Lagrangian object.

Field Table

.fr symbol	Particle	Spin	SU(3) rep	SU(2) rep	U(1) / charge	Self- conjugate	Mass
N1	N	1/2 Majorana fermion	singlet, no color index declared	singlet, no weak index declared	$Q = 0$; hy- percharge not declared	yes	$\mathbf{mN1}$, internal: $\frac{v^2}{\Lambda} C_{ee} + C_{e\mu} + C_{e\tau} + C_{\mu\mu} + C_{\mu\tau} + C_{\tau\tau} $

Width:

$$\Gamma_N = \mathbf{WN1} = 0.$$

PDG code:

9900012.

External Parameters

Symbol	Value	Multiplies / enters	Physical meaning
Lambda	200000.	Appears in the denominator of $\mathbf{mN1}$: $m_N \propto v^2/\Lambda$	Heavy mass scale suppressing the effective Majorana mass operator
Cee	1.1	Contribution to the sum defining $\mathbf{mN1}$	Dimensionless coefficient associated with the ee flavor entry
Cem	1.0	Contribution to the sum defining $\mathbf{mN1}$	Dimensionless coefficient associated with the $e\mu$ flavor entry
Cet	1.3	Contribution to the sum defining $\mathbf{mN1}$	Dimensionless coefficient associated with the $e\tau$ flavor entry
Cmm	1.4	Contribution to the sum defining $\mathbf{mN1}$	Dimensionless coefficient associated with the $\mu\mu$ flavor entry
Cmt	1.5	Contribution to the sum defining $\mathbf{mN1}$	Dimensionless coefficient associated with the $\mu\tau$ flavor entry
Ctt	1.6	Contribution to the sum defining $\mathbf{mN1}$	Dimensionless coefficient associated with the $\tau\tau$ flavor entry

The external coefficients do not appear as independent flavor-dependent vertices in the displayed interaction terms. In this file they enter the Lagrangian only through the internal mass parameter $\mathbf{mN1}$.

Physics Summary

The file adds one electrically neutral, self-conjugate Majorana fermion N , with mass generated by an effective scale Λ and dimensionless flavor coefficients C_{ij} . The new fermion couples left-chirally to SM charged leptons and neutrinos through $W^\pm$, Z , the Higgs boson, and the electroweak Goldstones in Feynman gauge. It mediates heavy neutral lepton production and decay channels such as $W^\pm \rightarrow N\ell^\pm$, $Z \rightarrow N\nu$, Higgs/Goldstone-associated interactions, and lepton-number-violating Goldstone-sector contact interactions tied to the Majorana mass.

Paper cross-check

Comparison of reconstruction.md Against the Paper Model

Located Model Definitions

The paper defines the underlying dimension-five SMEFT model in “**The Standard Model at Dimension Five**”, with the Weinberg operator in **Eq. (2)**,

$$\mathcal{L}_5 = \frac{C_5^{\ell\ell'}}{\Lambda} [\Phi \cdot L_\ell^c] [L_{\ell'} \cdot \Phi] + \text{H.c.},$$

the effective Majorana mass in **Eq. (3)**,

$$m_{\ell\ell'} = C_5^{\ell\ell'} v^2 / \Lambda,$$

and the EFT Lagrangian in **Eq. (4)**. In unitary gauge, the post-EWSB Weinberg terms are written in **Eq. (5)**. The collider prescription replaces the $(\nu_\ell \nu_{\ell'})$ current by an unphysical Majorana fermion N , with the charged-current interaction in **Eq. (8)** and the implementation mass prescription in **Eq. (9)**.

The implementation-level field and interaction definitions are in the **Appendix: Technical details on methodology**. The generic-gauge Higgs/Goldstone definitions are in **Eqs. (16)-(17)**, the expanded Weinberg operator is in **Eqs. (18)-(22)**, the SMWeinberg UFO interaction Lagrangian is in **Eqs. (23)-(29)**, and the UFO parameter table is **Table II**.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
SMEFT Lagrangian $\mathcal{L}_{\text{SMEFT}} = \mathcal{L}_{\text{SM}} + \mathcal{L}_5 + \mathcal{O}(\Lambda^{-2})$, Eq. (4)	LFull = LSM + LD5	agree	The reconstruction mirrors the paper’s structure: SM plus the dimension-five/new-sector implementation.
Weinberg operator $C_5^{\ell\ell'} [\Phi \cdot L_\ell^c] [L_{\ell'} \cdot \Phi] / \Lambda + \text{H.c.}$, Eq. (2)	Replaced by N -mediated LD5 terms	agree	The reconstruction does not keep the gauge-invariant operator literally; this matches the paper’s stated UFO prescription that models the current with an unphysical Majorana N .
Effective mass $m_{\ell\ell'} = C_5^{\ell\ell'} v^2 / \Lambda$, Eq. (3)	$m_N = (v^2 / \Lambda) C_{ee} + C_{e\mu} + C_{e\tau} + C_{\mu\mu} + C_{\mu\tau} + C_{\tau\tau} $	agree	This matches the paper’s implementation prescription in Eq. (9), not a single flavor-entry mass.
Unphysical Majorana neutrino N with mass m_N , text before Eq. (9), Table II	Field table: $N1$, Majorana, self-conjugate, neutral, mass mN1	agree	The paper calls N an unphysical Majorana neutrino and lists m_N as the internal mass parameter. Neutral singlet status is inferred consistently from the implementation.

paper term (eq. ref)	reconstruction term	verdict	notes
$m_N = C_{ee}^5 + C_{e\mu}^5 + C_{e\tau}^5 + C_{\mu\mu}^5 + C_{\mu\tau}^5 + C_{\tau\tau}^5 v^2/\Lambda$, Eq. (9)	Same expression with Cee, Cem, Cet, Cmm, Cmt, Ctt	agree	Same six independent symmetric flavor coefficients enter only through m_N .
Γ_N , Table II	$\Gamma_N = \text{WN1} = 0$	agree	Table II identifies the width parameter; the reconstruction's zero-width value is compatible with an internal unphysical mediator used for the prescription.
Generic-gauge Higgs doublet $\sqrt{2}\Phi = (-i\sqrt{2}G^+, v+h+iG^0)^T$, Eqs. (16)-(17)	Uses $H, G^0, G^-/G^+$ Goldstones in LNHX, LNGX, LGGX	agree	The reconstruction uses the charge-conjugate Goldstone orientation for several “bare” terms, then adds H.c.; physics content matches.
Weinberg post-EWSB neutrino mass term $-C_5^{\ell\ell'} v^2 \nu_\ell^c \nu_{\ell'}/(2\Lambda) + \text{H.c.}$, Eq. (5), Eq. (18)	$-\frac{1}{2}m_N \bar{N}N$ in LNKin	agree	This is the N -mediated implementation of the mass insertion/current, with the implementation mass of Eq. (9).
Single-Higgs Weinberg term $-C_5^{\ell\ell'} v h \nu_\ell^c \nu_{\ell'}/\Lambda + \text{H.c.}$, Eq. (5), Eq. (18)	$-g_W m_N H/(2M_W) \sum_i (\bar{N} P_L \nu_i + \bar{\nu}_i P_R N)$ in LNHX	agree	Using $M_W = g_W v/2$, the coefficient is $-m_N/v$, matching the implementation form in Eq. (25).
Double-Higgs Weinberg term $-C_5^{\ell\ell'} h h \nu_\ell^c \nu_{\ell'}/(2\Lambda) + \text{H.c.}$, Eq. (5), Eq. (18)	$-g_W^2 m_N H^2/(8M_W^2) \sum_i (\bar{N} P_L \nu_i + \bar{\nu}_i P_R N)$ in LNHX	agree	Coefficient equals $-m_N/(2v^2)$, matching Eq. (25).
Charged current $-g_W W_\mu^+ \sum_\ell \bar{N} \gamma^\mu P_L \ell^-/\sqrt{2} + \text{H.c.}$, Eq. (8), Eq. (23)	$+g_W W_\mu^+ \sum_i \bar{N} \gamma^\mu P_L \ell_i/\sqrt{2} + \text{H.c.}$ in LNCC	disagree	Field content, chirality, flavor sum, and H.c. structure match, but the overall sign is opposite to Eqs. (8) and (23).
Neutral current $-g_W Z_\mu \sum_\ell \bar{N} \gamma^\mu P_L \nu_\ell/(2 \cos \theta_W) + \text{H.c.}$, Eq. (24)	$+g_W Z_\mu \sum_i (\bar{N} \gamma^\mu P_L \nu_i + \bar{\nu}_i \gamma^\mu P_L N)/(2 \cos \theta_W)$ in LNNC	disagree	Same field content and chirality, but the coefficient sign is opposite to Eq. (24).
Higgs interaction $-g_W m_N h(1 + g_W h/(4m_W)) \sum_\ell \bar{N} P_L \nu_\ell/(2m_W) + \text{H.c.}$, Eq. (25)	Same coefficient and H.c. in LNHX	agree	Matches coefficient, Higgs expansion, chirality, and flavor-universal implementation.

paper term (eq. ref)	reconstruction term	verdict	notes
Charged-Goldstone term $-ig_W m_N G^+(1 + g_W h/(2m_W)) \sum_\ell (\bar{N} P_L \ell + \bar{\ell}^c P_L N)/(2\sqrt{2}m_W) + \text{H.c.}, \text{Eq. (26)}$	$+ig_W m_N G^-(1 + g_W H/(2M_W)) \sum_i (\bar{\ell}_i P_R N + \bar{N} P_R \ell_i^c)/(2\sqrt{2}M_W) + \text{H.c. in LNGX}$	agree	Reconstruction writes the charge-conjugate G^- piece as the bare term; with H.c., it matches Eq. (26). The P_R form is the conjugate bilinear of the paper's P_L form.
Neutral-Goldstone term $-ig_W m_N G^0(1 + g_W h/(2m_W)) \sum_\ell (\bar{N} P_L \nu_\ell + \bar{\nu}_\ell^c P_L N)/(2m_W) + \text{H.c.}, \text{Eq. (27)}$	$+ig_W m_N G^0(1 + g_W H/(2M_W)) \sum_i \bar{\nu}_i P_R N/(2M_W) + \text{H.c. in LNGX}$	agree	Reconstruction displays the Hermitian-conjugate bilinear as the bare term; with H.c., coefficient and chirality match Eq. (27).
$G^+ G^+$ contact term $+g_W^2 m_N (G^+)^2 \sum_{\ell, \ell'} \bar{\ell}^c P_L \ell'/(8m_W^2) + \text{H.c. in LNGGX}$ H.c., Eq. (28)	$+g_W^2 m_N (G^-)^2 \sum_{i, j} \bar{\ell}_i P_R \ell_j^c/(4M_W^2) + \text{H.c. in LNGGX}$	agree	Reconstruction writes the H.c. $G^- G^-$ orientation; coefficient and all ordered flavor combinations match.
$G^0 G^0$ contact term $+g_W^2 m_N (G^0)^2 \sum_\ell \bar{N} P_L \nu_\ell/(8m_W^2) + \text{H.c. in LNGGX}$ H.c., Eq. (28)	$+g_W^2 m_N (G^0)^2 \sum_i \bar{\nu}_i P_R N/(8M_W^2) + \text{H.c. in LNGGX}$	agree	Reconstruction writes the conjugate bilinear; H.c. completes the paper term.
Mixed $G^0 G^+$ contact term $+g_W^2 m_N G^- G^0 \sum_i (\bar{\ell}_i P_R N + \bar{N} P_R \ell_i^c)/(4\sqrt{2}M_W^2) + \text{H.c. in LNGGX}$ $+g_W^2 m_N G^0 G^+ \sum_\ell (\bar{N} P_L \ell + \bar{\ell}^c P_L N)/(4\sqrt{2}m_W^2) + \text{H.c.}, \text{Eq. (29)}$	$+g_W^2 m_N G^- G^0 \sum_i (\bar{\ell}_i P_R N + \bar{N} P_R \ell_i^c)/(4\sqrt{2}M_W^2) + \text{H.c. in LNGGX}$	agree	Reconstruction gives the charge-conjugate $G^- G^0$ bare orientation; full H.c. agrees with Eq. (29).
Direct flavor-dependent vertices proportional to individual $C_5^{\ell\ell'}$, Eqs. (18)-(22)	No independent C_{ij} -dependent vertices; coefficients enter only through m_N	agree	This matches the SMWeinberg UFO prescription described around Eq. (9), although it is not the literal flavor-basis Weinberg operator.
Z , Higgs, and Goldstone interactions in the appendix, Eqs. (24)-(29)	Included in LNNC, LNHX, LNGX, LNGGX	agree	These terms would look extra if compared only to the main-text Eq. (8), but they are explicitly part of the appendix UFO Lagrangian.
N kinetic term	$\frac{i}{2} \bar{N} \gamma^\mu \partial_\mu N$ in LNKin	agree	The paper does not print this term explicitly, but a single free Majorana field with mass m_N is required by the implementation. Ordinary derivative is consistent with N being neutral/singlet.

paper term (eq. ref)	reconstruction term	verdict	notes
Gauge-covariant kinetic coupling of N	None	agree	The paper’s N is an unphysical Majorana neutrino with prescribed EW interactions, not a field carrying SM gauge indices.

Disagreements to Check

1. **Charged-current overall sign, Eqs. (8) and (23)** — severity: **convention**.

Human check: compare the original FeynRules/UFO vertex convention for $W^+N\ell^-$ against the paper’s printed $\Delta\mathcal{L}$, because the reconstruction has the same fields and chirality but the opposite Lagrangian sign.

2. **Neutral-current overall sign, Eq. (24)** — severity: **convention**.

Human check: verify whether the $ZN\nu$ sign in the implementation is tied to the same convention as the charged-current sign or whether it is a transcription/sign error relative to the appendix.

No substantive mismatch was found in the mass prescription, flavor sums, Majorana nature of N , Higgs terms, Goldstone terms, Hermitian-conjugate structure, or chirality assignments after accounting for charge-conjugate bilinears.

Overall Assessment

The reconstruction closely matches the SMWeinberg implementation described in the paper’s appendix: it identifies the same unphysical neutral Majorana mediator N , the same internally calculated mass m_N , the same flavor-universal $W/Z/H$ and Goldstone interactions, and the same replacement of explicit $C_5^{\ell\ell'}$ -dependent Weinberg vertices by coefficients entering through m_N . The main caveat is an apparent overall sign difference in the reconstructed W and Z interaction terms relative to the printed Eqs. (8), (23), and (24); this is likely convention-level but should be checked against the actual FeynRules/UFO vertex output before relying on relative signs in amplitudes involving interference.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field’s representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 9 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model’s identity.

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: [] approve [] approve with corrections [] reject
- Notes: